NORADRENALINE

High-alert drug

Central line — extravasation

CHEMICAL NAME (R)-4-(2-amino-1-hydroxyethyl)benzene-1,2-diol (as tartrate); also called norepinephrine

CLASS Sympathomimetic catecholamine vasopressor (predominantly α1)

TRADE NAMES Levophed · Noradrenaline acid tartrate · Norepinephrine bitartrate

THERAPEUTIC USES / INDICATIONS

- Septic shock; first-line vasopressor.
- Other forms of vasodilatory shock and severe hypotension unresponsive to fluids.
- Cardiogenic shock with low BP (with inotrope).

ADMINISTRATION

- **IV infusion** via central venous line with dedicated lumen and syringe pump (peripheral only short-term via large vein with close monitoring).
- Dilute in 5% dextrose (or per local protocol); label clearly as base or tartrate.
- Never bolus or flush the line — sudden surge in BP.
- Arterial line for continuous BP monitoring.

DOSAGES

- **Adult:** start 0.05–0.1 mcg/kg/min, titrate every 5–15 min to MAP ≥65 mmHg (usual 0.05–1 mcg/kg/min).
- **Children:** 0.05–2 mcg/kg/min.
- Correct hypovolaemia first (fluids) where possible.
- Wean gradually — do not stop abruptly.

PHARMACOKINETICS

- **Onset:** 1–2 min. **Duration:** 1–2 min after stopping.
- Rapidly metabolised by COMT and MAO.
- Metabolites excreted in urine.

PHARMACODYNAMICS

- Potent **α1 agonist** → vasoconstriction → ↑ systemic vascular resistance and MAP.
- Moderate **β1** → ↑ contractility; minimal β2.
- Reflex bradycardia may occur.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, anxiety.
- Reflex bradycardia.
- Cold extremities.

ADVERSE EFFECTS (SERIOUS)

- Extravasation → tissue necrosis (antidote: phentolamine infiltration).
- Peripheral, mesenteric and renal ischaemia; digital gangrene.
- Arrhythmias; hypertension.
- Myocardial ischaemia.

CONTRAINDICATIONS & CAUTIONS

- Uncorrected hypovolaemia (relative).
- Mesenteric or peripheral vascular thrombosis (unless life-saving).
- **Caution:** use of MAOIs, halothane anaesthesia.

KEY INTERACTIONS

- **MAOIs, TCAs, linezolid:** severe hypertension.
- **Volatile anaesthetics:** arrhythmias.
- **β-blockers:** unopposed α → hypertension, bradycardia.

NURSING MANAGEMENT

- Continuous arterial BP, ECG, SpO_2; MAP every 5–15 min during titration.
- Check central line site hourly; peripheral site every 15–30 min if used.
- Monitor perfusion: urine output, lactate, peripheries, capillary refill.
- Prepare next syringe in advance; no interruption during syringe change (double-pumping).
- Have phentolamine available.

PATIENT EDUCATION

- Explain to family that the infusion supports blood pressure and is adjusted continuously.

KEY POINTS

- α1 vasopressor — first-line in septic shock.
- Central line; never bolus or flush.
- Target MAP ≥65 mmHg.
- Extravasation → phentolamine.